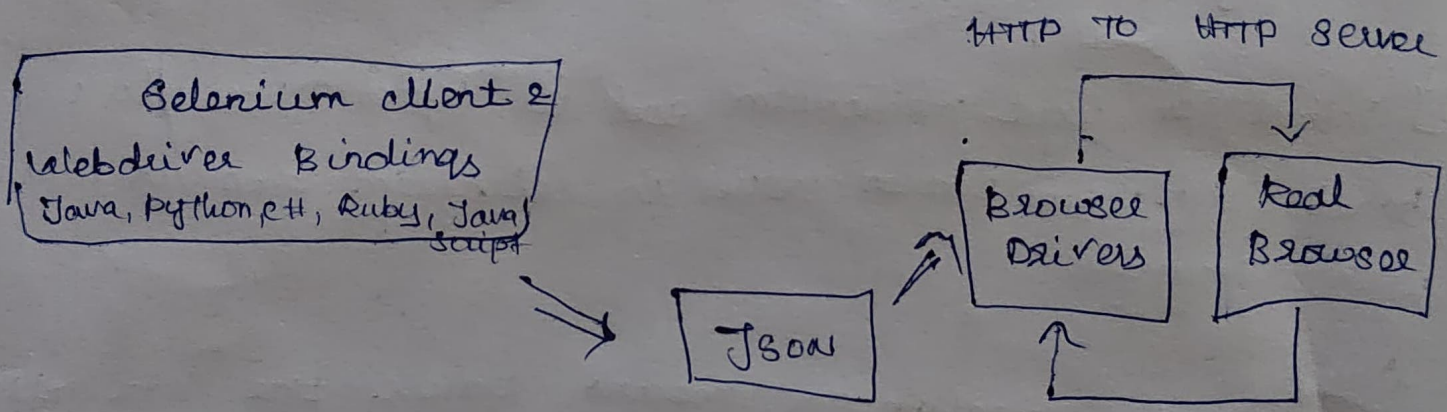
II Explain the Selenium architecture with diagram. Explain with example program to open a website using driver.

Ans: Selenium architecture explains a Selenium interact with different browsers to automate web applications. It shows a test Script how to communicates with the browser through webdriver and browser drivers.



Selenium client :

* These are programming Language used to write test automation Scripts.

JSON :

* This is protocol. used to communication between Selenium and the browser driver.

Browser drivers :

* Each browser have a own driver.

Real Browser :

* Real Browser used to execute the actual browser commands.

Program :

```
Public class OpenWebsite {
    Public Static Void main (String [] args) {
        WebDriver driver = new ChromeDriver();
        driver.get("https://www.Amazon.com");
        driver.manage().window().maximize();
        driver.quit();
    }
}
```

2] What is wait mechanism and its types? How do u handle synchronization issues in selenium explain with example.

Ans:

Wait mechanism means pause the execution of the script until the specific condition met.

Types: Implicit wait
Explicit wait
Fluent wait

Implicit wait: This wait applies globally for the entire script.

Explicit wait: this wait applies specific condition.

Fluent wait: This is custom polling.

Synchronization issue:

Synchronization issue occurs when Selenium executes faster than the web application.

Example:

(i) Selenium clicks login button
(ii) Alert page loads slowly
(iii) Element not loaded - error will be occur. So, these kind of issues can use the explicit wait to fix the problem.

Example:

```
Public class Explicitwait {
    Public Static Void main (String [] args){
    WebDriver driver = new ChromeDriver ();
    driver. get ("https://www.amazon.com");
    WebDriverWait wait = new WebDriverWait
                (driver, Duration. Of Seconds
                (10))
    wait. until ( Expected conditions.
            VisiblityOf ElementLocated(By.name("q")));
    driver. findElement (By.name("q")). sendKeys("Selenium
    driver. quit ();
    }
}
```

3] How do you take page and element screenshots in Selenium? Explain with example.

Ans:

In Selenium a screenshot used to capture the current browser screen during the test execution. The screenshot mostly covers the

* Debugging test failures
* capturing error message
* Report the test result.

Usually, TakesScreenshot interface used to take the Selenium screenshot.

Types: (i) page screenshot
(ii) element screenshot

(i) page screenshot

* This is used to take the full page of the screenshot.

Example:

```
public class PageScreenshot {
    Public static void main(String [] args) {
    TakesDriver drive = new ChromeDriver();
    driver.get("https://www.amazon.com");
    File Src = (TakesScreenshot) drive).
        getScreenshotAS(outputType.FILE);
    File des = new File("c:\\Screenshots\\google.
                                    png");
```

```
FileUtils.copyFile(src, dest);
driver.quit();
}
}
```

4) What is the diff between find Element and find Elements. Explain with example:

findElement: return the single element, if doesn't find any elements, it throws the exception (NO Such element Exception)

findElements: returns the list of elements. If doesn't find any elements, it throws the empty list.

Example for findElement:

```
Public class findElement {
    public static void main(String[] args) {

        WebDriver driver = new ChromeDriver();
        driver.get("https://www.amazon.com");
        WebElemen element = driver.findElement
                (By.id("q"));
        element.sendKeys("selenium webdriver");
        driver.quit();
    }
}
```

Example of FindElements:

```
public class FindElements {
    public static void main (String[] args) {
WebDriver driver = new ChromeDriver();
driver.get("http://www.amazon.com");
List<WebElement> element = driver.
        findElements(By.tagName("a"));
System.out.println("total link:" + element.
                                size());
driver.quit();
    }
}
```

5] What are Relative Locators? Explain its types with example?

Relative locators are used to locate the elements based on the position of the elements in the page. Instead of writing complex xpath, you can locate elements using their position like left & right.

types of Relative locators:

* above()
* below()

* toLeftOf()
* toRightOff()
* near()

(i) above():

above() used to find element in above another element on page.

Example;

```
WebElement pwd = driver.findElement(By.id("password"));
WebElement username = driver.findElement(with
        (By.tagName("input")).above(password));
username.sendKeys("Revathi");
```

(ii) below();

below() used to find an element that is below another element.

Example:

```
WebElement username = driver.findElement(By.
        id("username"));
WebElement password = driver.findElement(with
        (By.tagName("input")).below(username));
password.sendKeys("123456");
```

(iii) toLeftOf():

toLeftOf() used to finds an element to left side of another element.

Example:

```
WebElement passwordbox = driver.FindElement(By.
                id("password"));

WebElement usernamebox = driver.FindElement(with
        (By.tagName("input")).toLeftOf(passwordbox));
```

(iii) toRightOf():

toRightOf() used to finds element to right side of another element.

Example:

```
WebElement username = driver.FindElement(By.id("
                username"));

WebElement remember = driver.FindElement(with(By.
        tagname("input")).toRightOf(username));
```

(iv) near()

near() finds an element o the ~~nigh~~ close ~~side of~~ the ~~oth~~ other element.

example:

```
WebElement label = driver.FindElement(By.id("email"));

WebElement email = driver.FindElement(with(By.
        tagname("input")).near(label));

emailTextbox.SendKeys("test");
```